

CLOUD COMPUTING

LAB NO # 08



SUBMITTED TO:

Sir Shohaib

SUBMITTED BY:

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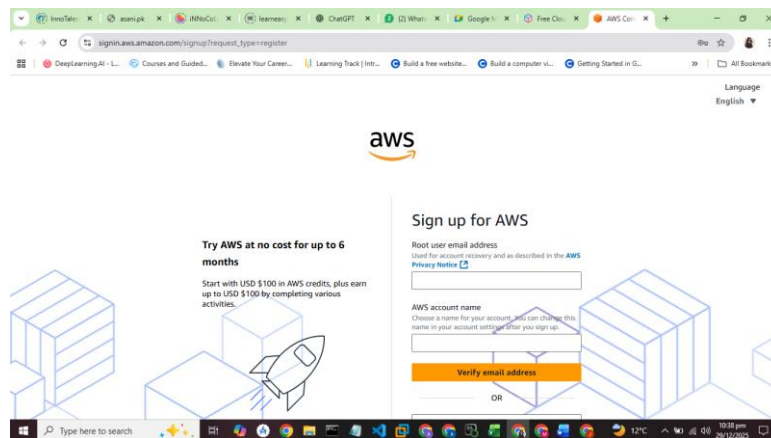
CLOUD COMPUTING

LAB NO # 08

Task 1: Create an AWS account and enable UAE (me-central-1)

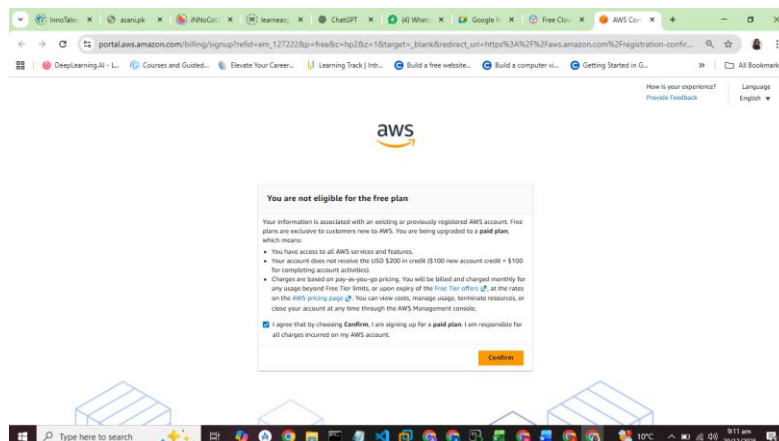
Step 1: Open the AWS Signup page

1. Open browser
2. Go to: <https://aws.amazon.com/resources/create-account/>



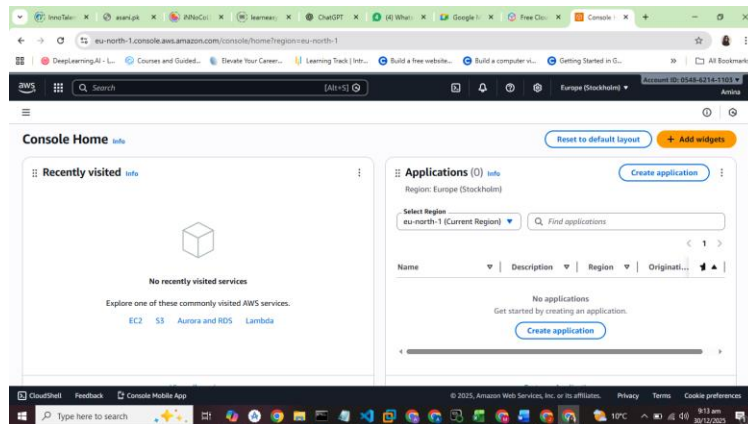
Step 2: Complete Registration

- Account type: Personal
- Plan: AWS Paid Plan
- Enter:
 - ✓ Contact details
 - ✓ Billing info (credit card)
 - ✓ Phone verification
- When you see account creation success or payment confirmation:



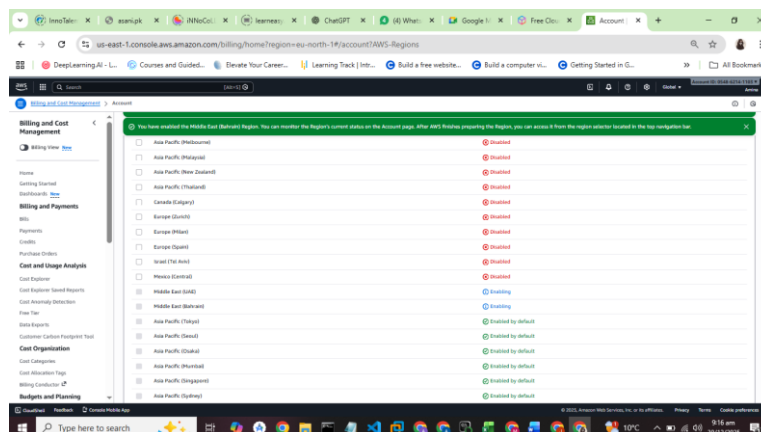
Step 3: Sign in using Root Account

- Login using root email (same used during signup)
- Once inside AWS Console Home:



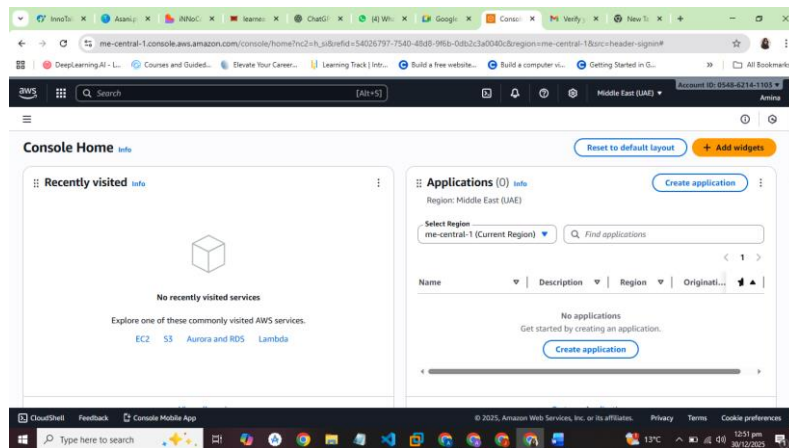
Step 4: Enable UAE Region

- Go to Region selector (top-right menu)
- Enable Middle East (UAE) — me-central-1
- Switch to me-central-1 region



Step 5: Final Summary Screenshot

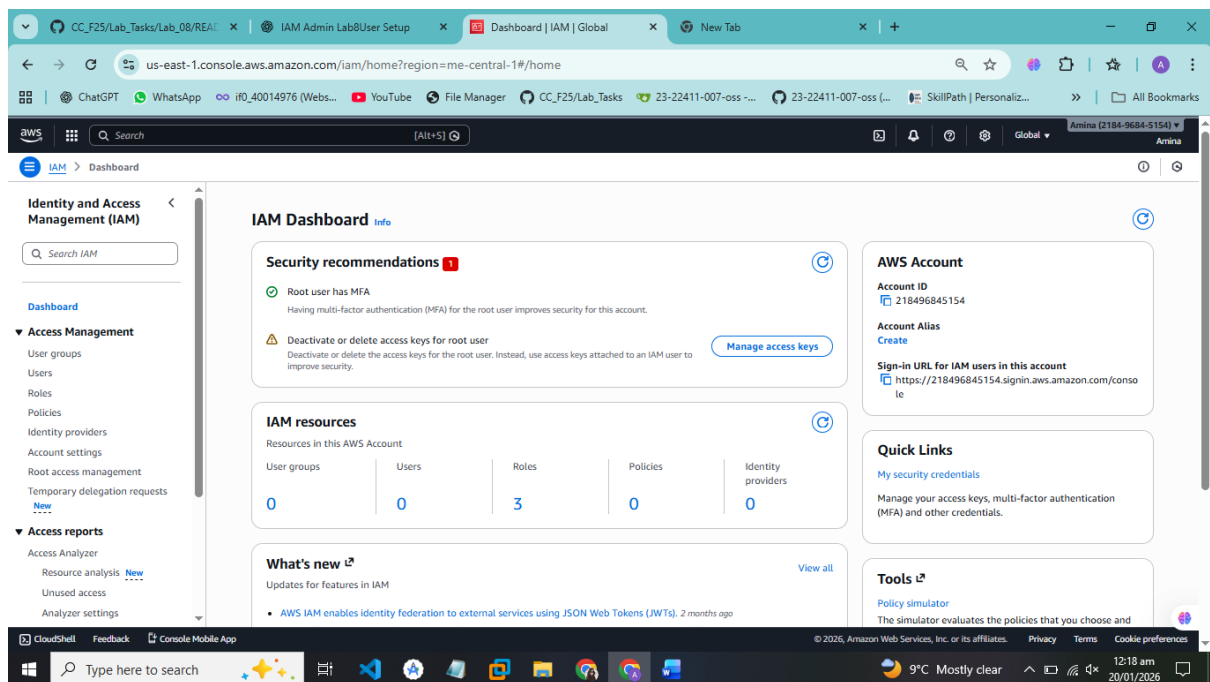
- AWS Console Home
- Must show:
 - ✓ Root email
 - ✓ Region = me-central-1



TASK 2 – IAM USERS (Admin + Lab8User)

Step 1: Open IAM

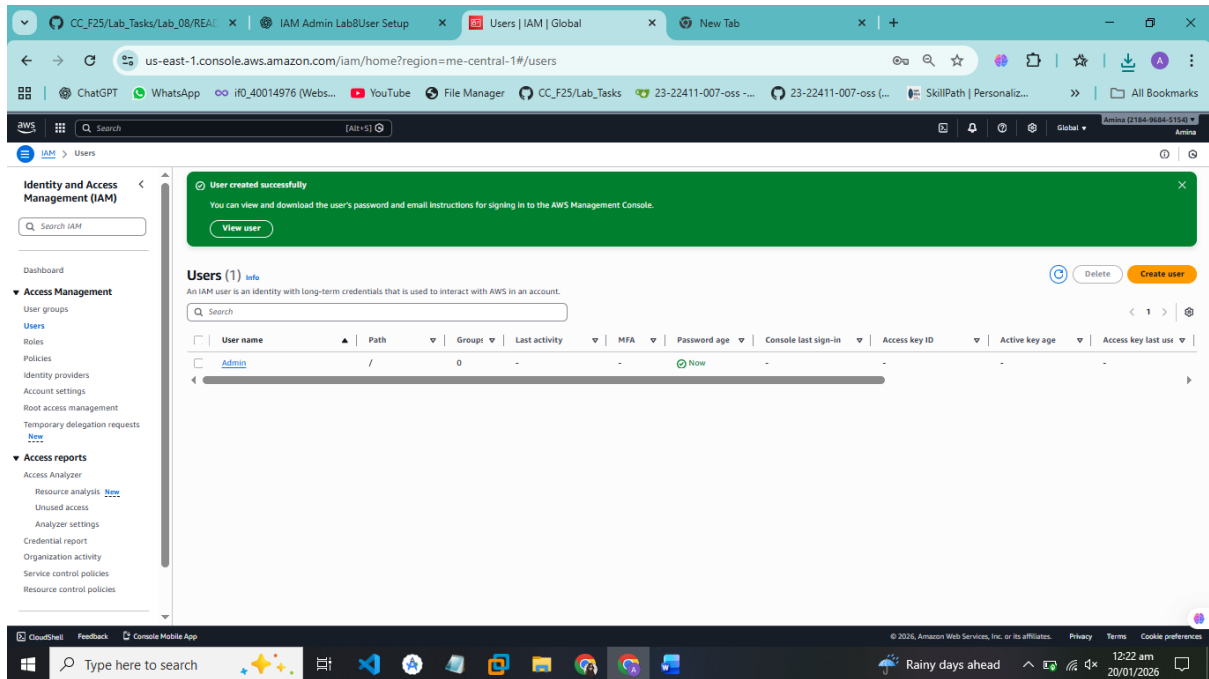
1. In AWS search bar type: IAM
2. Open IAM service



Step 2: Create Admin user

1. IAM → Users → Create user
2. Username: Admin
3. Tick:
 - ✓ AWS Management Console access
4. Password:

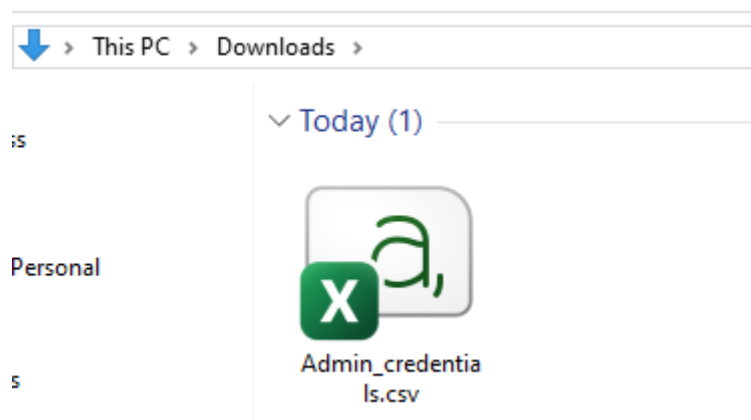
- Autogenerate OR Custom
5. Permissions:
- Attach policies directly
 - Select AdministratorAccess
6. Create user



task2_admin_create_confirmation.png

Step 3: Download Admin CSV

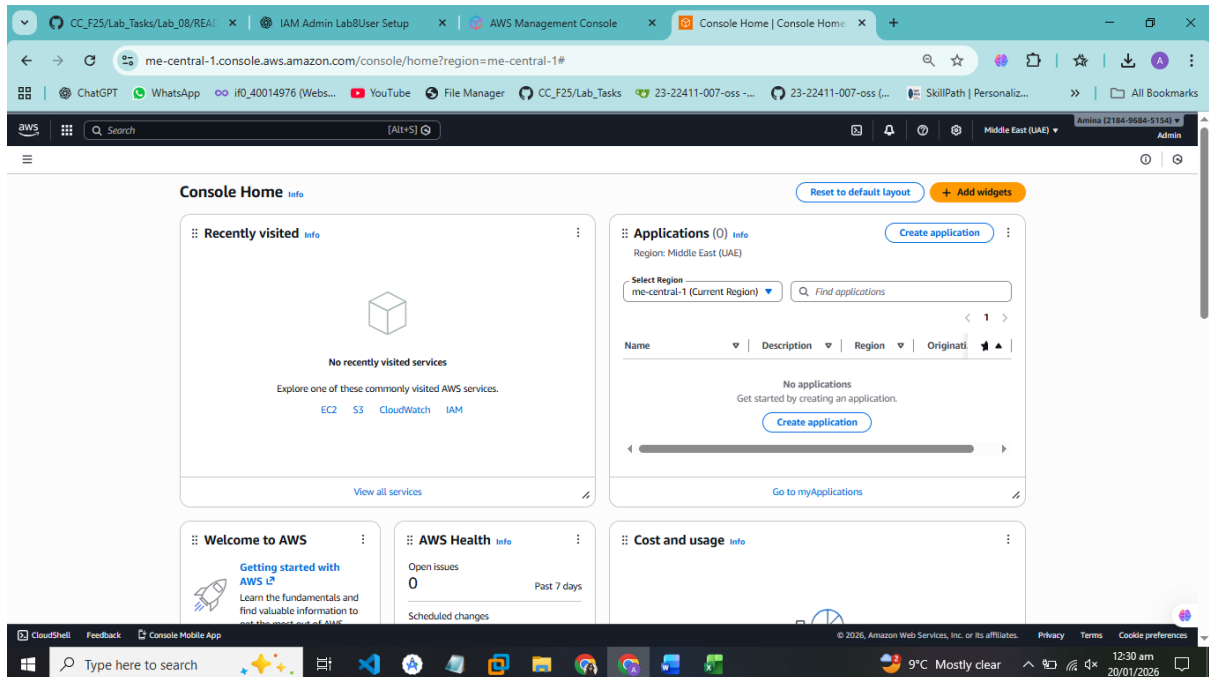
1. Click Download .csv
2. Save it to Desktop
3. Open File Explorer (DON'T open CSV)



task2_admin_csv_and_signin_url.png

Step 4: Login as Admin

1. Logout root
2. Open browser → use Sign-in URL from CSV
3. Login as Admin



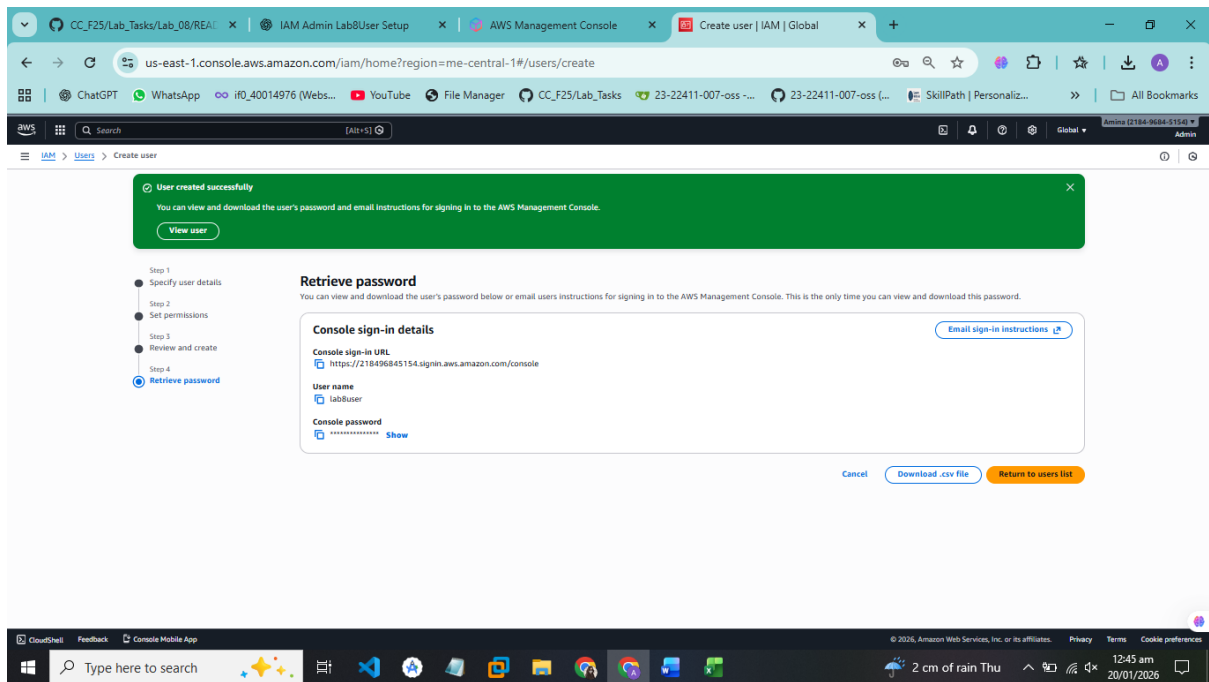
 Screenshot Admin console

Save as:

task2_admin_console_after_login.png

Step 5: Create Lab8User

1. IAM → Users → Create user
2. Username: Lab8User
3. Console access → YES
4. Permissions:
 - AdministratorAccess
5. Create user + download CSV



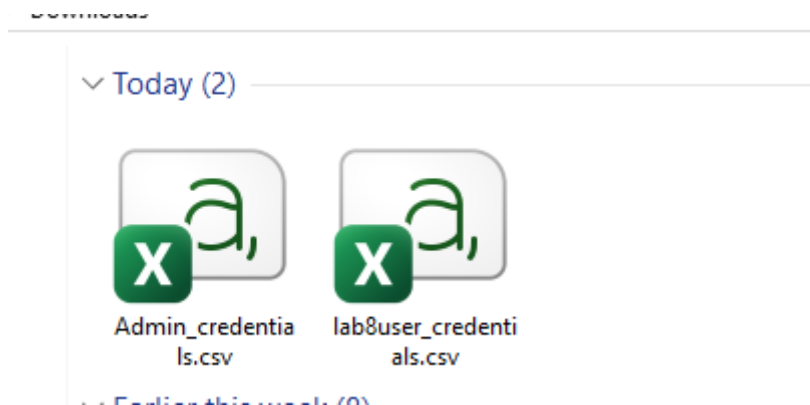
 Screenshot success page

Save as:

task2_create_lab8user_and_csv.png

Step 6: Save Lab8User CSV

1. Open File Explorer
2. Show CSV file only



 Screenshot

Save as:

task2_lab8user_csv_saved.png

Step 7: Login as Lab8User

1. Logout Admin
2. Login using Lab8User URL

	E5											
	A	B	C	D	E	F	G	H	I	J	K	L
1	User name	Path	Console sign-in URL									
2	Lab8User	/	https://941618064465.signin.aws.amazon.com/console									

 Screenshot console

Save as:

task2_lab8user_logged_in.png

Step 8: Task 2 Summary

1. IAM → Users
2. Show:
 - Admin
 - Lab8User

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Acc
☐	Admin	/	0	✓ 14 minutes ago	-	✓ 13 minutes	✓ 14 minutes ago	-
☐	Lab8User	/	0	✓ 3 minutes ago	-	✓ 2 minutes	✓ 3 minutes ago	-

 Screenshot

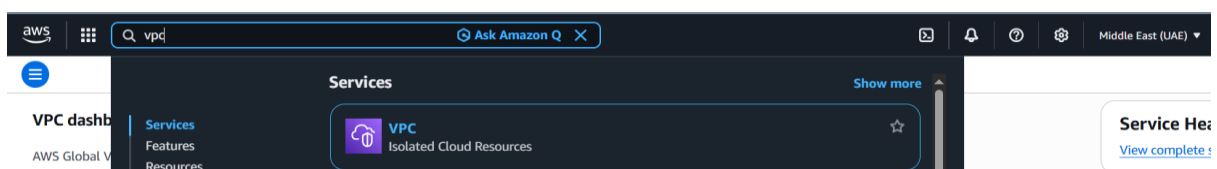
Save as:

task2_summary.png

TASK 3 – VPC INVENTORY

Step 1: Open VPC

1. Search: VPC



 Screenshot

Save as:

task3_open_vpc_console.png

Step 2: VPCs

1. Left menu → VPCs

Your VPCs

VPCs

VPC encryption controls

Your VPCs (1) Info

Last updated 3 minutes ago

Actions

Create VPC

Find VPCs by attribute or tag

< 1 >

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4
☐	-	vpc-05883252ecb09ac3e	Available	-	-	Off	172.3

 Screenshot

Save as:

task3_vpcs_list.png

Step 3: Subnets

1. Left menu → Subnets

Subnets (3) [Info](#)

Last updated 2 minutes ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	-	subnet-015aa793c89df0a82	Available	vpc-05883252ecb09ac3e	Off	172.31.32.0/20
☐	-	subnet-084c39d400b49a9b4	Available	vpc-05883252ecb09ac3e	Off	172.31.16.0/20
☐	-	subnet-0336f1ba99f13aae0	Available	vpc-05883252ecb09ac3e	Off	172.31.0.0/20

 Screenshot

Save as:

task3_subnets_list.png

Step 4: Route Tables

Network ACLs (1) [Info](#)

[Actions](#) [Create network ACL](#)

Find Network ACLs by attribute or tag

☐	Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
☐	-	acl-0126863aa4fc9be5f	3 Subnets	Yes	vpc-05883252ecb09ac3e	2 Inbound

 Save as:

task3_route_tables_list.png

Step 5: Network ACLs

Network ACLs (1) [Info](#)

[Actions](#) [Create network ACL](#)

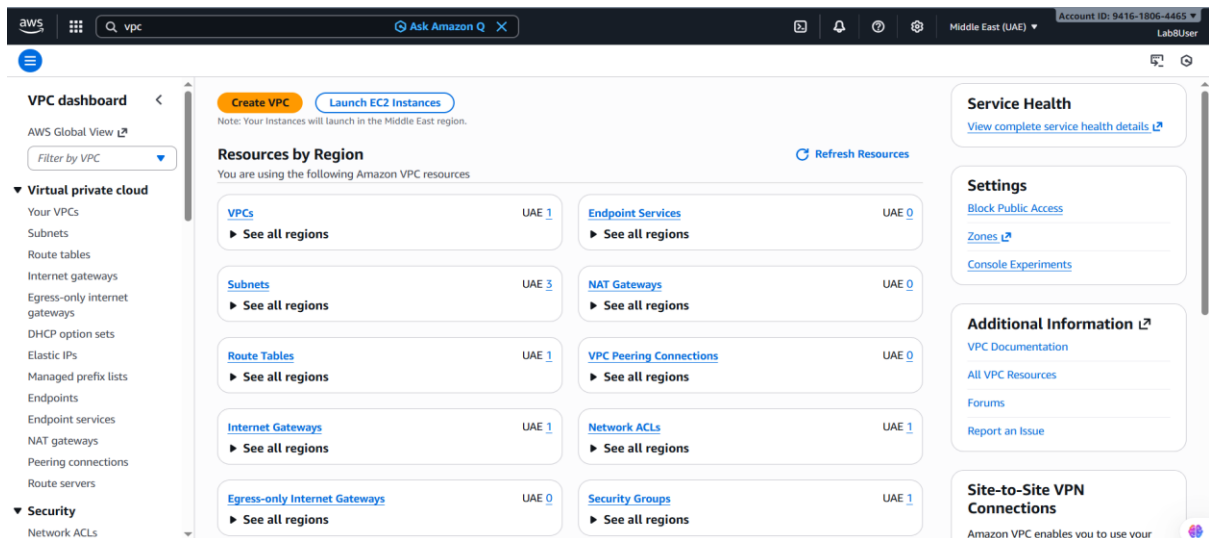
Find Network ACLs by attribute or tag

☐	Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
☐	-	acl-0126863aa4fc9be5f	3 Subnets	Yes	vpc-05883252ecb09ac3e	2 Inbound

 Save as:

task3_network_acls_list.png

Step 6: Task 3 Summary



 Combined screenshot

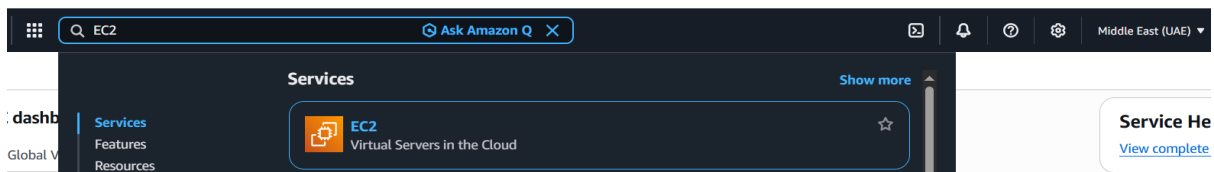
Save as:

task3_summary.png

TASK 4 – EC2 + DOCKER + GITEA

Step 1: Open EC2

Search: EC2



 Save as:

task4_open_ec2_console.png

Step 2: Launch Instance

Click Launch instance

Setting	Value
Name	Lab8Machine
AMI	Amazon Linux 2
Type	t3.micro
Key pair	Create → Lab8Key → ED25519
Security group	Create new → SSH from My IP
Storage	Default

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - *required*

Lab8SecurityGroup

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./()#,@[]+=&:{}!\$*

Description - *required* [Info](#)

launch-wizard-2 created 2025-11-20T20:28:24.713Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 223.123.91.51/32)

Remove

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

My IP

Name [Info](#)

Q Add CIDR, prefix list or security group

223.123.91.51/32 X

Description - *optional* [Info](#)

e.g. SSH for admin desktop

Review page

Save as:

task4_launch_instance_config.png

↓ > This PC > Downloads				
	Name	Date modified	Type	Size
▼ Today (3)				
	Lab8Key.pem	12/29/2025 2:08 PM	PEM File	1 KB

Show .pem file downloaded

Save as:

task4_keypair_download.png

Step 3: Instance running

1. EC2 → Instances
2. Wait until Running
3. Copy Public IPv4

Instance summary for i-0bda6824dcef7f953 (Lab8Machine) [Info](#)

Updated less than a minute ago

Instance ID

i-0bda6824dcef7f953

IPv6 address

–

Hostname type

IP name: ip-172-31-3-63.me-central-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

3.28.60.106 [Public IP]

IAM Role

–

IMDSv2

Required

Operator

–

Public IPv4 address

3.28.60.106 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-3-63.me-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-05883252ecb09ac3e

Subnet ID

subnet-0336f1ba99f13aae0

Instance ARN

arn:aws:ec2:me-central-1:941618064465:instance/i-Obda6824dcef7f953



Connect

Instance state ▼

Actions ▼

Private IPv4 addresses

172.31.3.63

Public DNS

ec2-3-28-60-106.me-central-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name

–

Managed

false



Save as:

task4_instance_running_console.png

Step 4: SSH from Windows

Open PowerShell

```
ssh -i "C:\Path\Lab8Key.pem" ec2-user@PUBLIC_IP
```

[illegible]

Save as:

task4_ssh_from_windows_to_ec2.png

Step 5: Install Docker + Compose

Run one by one:

```
sudo yum update -y
```

```
sudo yum install -y docker
```

```
sudo mkdir -p /usr/local/lib/docker/cli-plugins
```

```
sudo curl -SL https://github.com/docker/compose/releases/latest/download/docker-compose-  
linux-x86_64 -o /usr/local/lib/docker/cli-plugins/docker-compose
```

```
sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
```

```
sudo systemctl start docker
```

```

ec2-user@ip-172-31-3-63 ~$ sudo yum update -y
Amazon Linux 2023 Kernel livepatch repository
Last metadata expiration check: 0:00:01 ago on Mon Dec 29 10:13:21 2025.
Dependencies resolved.
Nothing to do.
Complete!

[ec2-user@ip-172-31-3-63 ~]$ sudo yum install -y docker
Last metadata expiration check: 0:37:06 ago on Mon Dec 29 10:13:21 2025.
Dependencies resolved.

```

Package	Architecture	Version	Repository	Size
Installing:				
docker	x86_64	25.0.13-1.amzn2023.0.2	amazonlinux	46 M
Installing dependencies:				
container-selinux	noarch	4:2.242.0-1.amzn2023	amazonlinux	58 k
containerd	x86_64	2.1.5-1.amzn2023.0.1	amazonlinux	23 M
iptables-libs	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	401 k
iptables-nft	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	183 k
libgroup	x86_64	3.0.1.amzn2023.0.1	amazonlinux	75 k
libnetfilter_conntrack	x86_64	1.0.8-2.amzn2023.0.2	amazonlinux	58 k
libnftnl	x86_64	1.0.1-19.amzn2023.0.2	amazonlinux	30 k
libnftnl	x86_64	1.2.2-2.amzn2023.0.2	amazonlinux	64 k
libg	x86_64	2.5-1.amzn2023.0.3	amazonlinux	83 k
runc	x86_64	1.3.3-2.amzn2023.0.1	amazonlinux	3.9 M
Transaction Summary				
Install 11 Packages				
Total download size: 74 M				
Installed size: 280 M				
Downloading Packages:				
(1/11): container-selinux-2.242.0-1.amzn2023.noarch.rpm			1.4 MB/s 58 kB	00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm			12 MB/s 401 kB	00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm			6.6 MB/s 183 kB	00:00
(4/11): libgroup-3.0.1.amzn2023.0.1.x86_64.rpm			2.3 MB/s 75 kB	00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm			1.8 MB/s 58 kB	00:00
(6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm			1.0 MB/s 30 kB	00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm			2.4 MB/s 84 kB	00:00
(8/11): libg-2.5-1.amzn2023.0.2.x86_64.rpm			2.6 MB/s 83 kB	00:00
(9/11): containerd-2.1.5-1.amzn2023.0.1.x86_64.rpm			64 MB/s 23 MB	00:00
(10/11): runc-1.3.3-2.amzn2023.0.1.x86_64.rpm			26 MB/s 3.9 MB	00:00
(11/11): docker-25.0.13-1.amzn2023.0.2.x86_64.rpm			67 MB/s 46 MB	00:00
Total			182 MB/s 74 MB	00:00

 Screenshot terminal

Save as:

task4_ec2_install_docker_compose_started.png

Step 6: Create compose.yaml

sudo vim compose.yaml

Paste content from:

<https://github.com/WaqasSaleem97/Gitea>

```
ec2-user@ip-172-31-3-63:~  
container_name gitea  
environment  
- DB_TYPE=postgres  
- DB_HOST=db:5432  
- DB_NAME=gitea  
- DB_USER=gitea  
- DB_PASSWD=gitea  
restart always  
volumes  
- gitea:/data  
ports  
- 3000:3000
```

 Screenshot vim

Save as:

task4_vim_compose_yaml_paste.png

Step 7: Verify file

ls -l

```
[ec2-user@ip-172-31-3-63 ~]$ ls  
compose.yaml
```

 Screenshot

Save as:

task4_compose_yaml_saved_ls.png

Step 8: Docker group

groups

sudo usermod -aG docker \$USER

exit

Reconnect and run:

Groups

```
PS C:\Windows\system32> ssh -i "D:\FJWU\Semester 5\Cloud Computing\CC Lab-Engr. Shoaib\Lab8\Lab8Key.pem" ec2-user@3.28.60.106
```

```
Last login: Mon Dec 29 10:11:52 2025 from 39.60.136.17
[ec2-user@ip-172-31-3-63 ~]$ groups
ec2-user adm wheel systemd-journal docker
```



Save as:

Step 9: Run Gitea

```
ec2-user@ip-172-31-3-63 ~]$ docker compose up -d
WARN[0000] /home/ec2-user/compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 16/18
[+] up 23/23 postgres:alpine [postgres:alpine] 112MB / 112MB Pulling
Image postgres:alpine Pulled
 0 1074353eec0d Pull complete
 0 51a9324a2bdc Pull complete
 0 1e827ac0fa1e Pull complete
 0 0629ac8b8cc5 Pull complete
 0 2e50a444bddf Pull complete
 0 4465107e1675 Pull complete
 0 cc2ee20b6816 Pull complete
 0 0dfd86289dd1 Pull complete
 0 98eccc0da868 Pull complete
 0 b747867e61cb Pull complete
Image gitea:gitea:latest Pulled
 0 2d35ebdd57d9 Pull complete
 0 9f4ae72c1f34 Pull complete
 0 4da94b3cc809 Pull complete
 0 8e016eca0bd3 Pull complete
 0 346e493cb6cb Pull complete
 0 95215379f1d1 Pull complete
Network ec2-user_webnet Created
Volume ec2-user_gitea_postgres Created
Volume ec2-user_gitea Created
Container gitea_db Created
Container gitea Created
```



Save as:

Step 10: Allow port 3000

Add:

[Edit inbound rules](#) [Info](#)

Edit Inbound Rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules info					
Security group rule ID	Type info	Protocol info	Port range info	Source info	Description - optional info
-	Custom TCP ▼	TCP	3000	Anywh... ▼	

 Screenshot

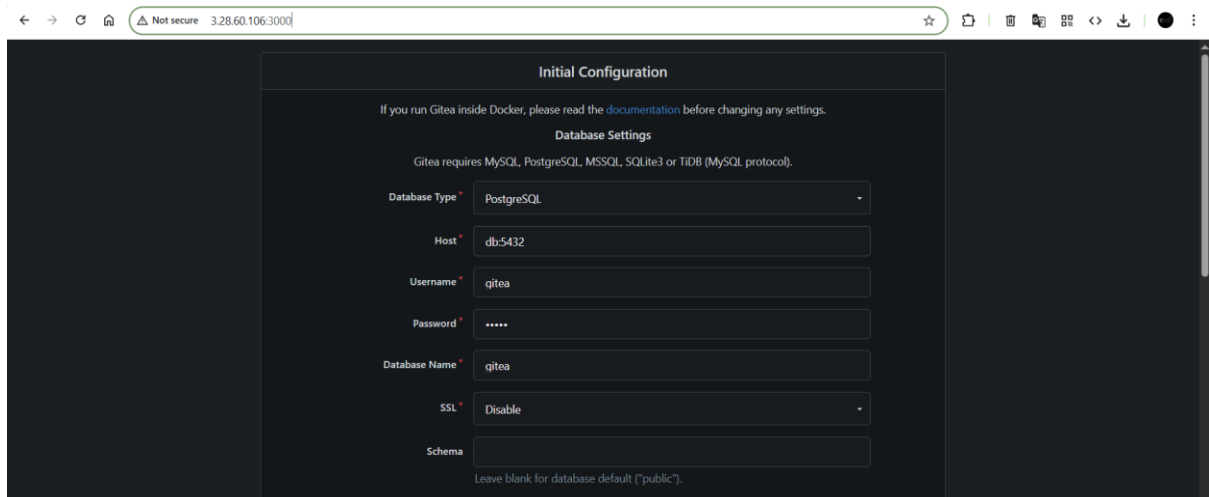
Save as:

task4_security_group_allow_3000.png

Step 11: Open Gitea

Browser:

http://PUBLIC_IP:3000

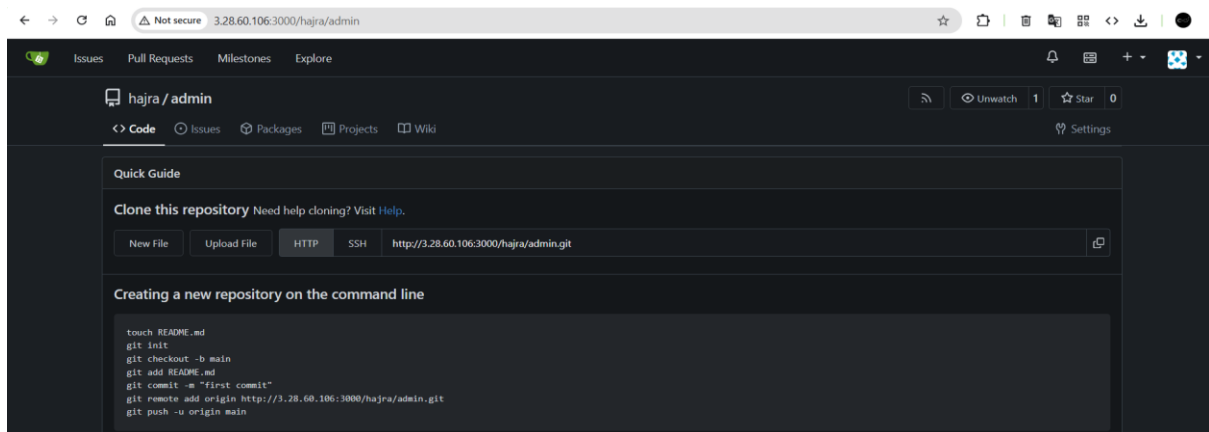


 Screenshot

Save as:

task4_gitea_install_page.png

Step 12: Create repo in Gitea



 Screenshot

Save as:

task4_gitea_create_repo.png

Step 13: Task 4 Summary

Instances (1) Info

Find Instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 D
Lab8Machine	i-0bda6824dcef7f953	Running	t3.micro	3/3 checks passed	View alarms	me-central-1c	ec2-3-28-60-11.compute.am

sg-052e06e3b019fbb66 - Lab8SecurityGroup

Details | **Inbound rules** | Outbound rules | Sharing | VPC associations | Tags

Inbound rules (1)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-08b66f74ccb44ad27	IPv4	Custom TCP	TCP	3000

📷 Combined screenshot

Save as:

task4_summary.png

CLEANUP (VERY IMPORTANT)

Terminate EC2 → delete volumes → delete SG → delete key → delete IAM users

📷 Screenshots:

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 D
Lab8Machine	i-0bda6824dcef7f953	Running	t3.micro	3/3 checks passed	me-central-1c	ec2-3-28-60-11.compute.am

Stop instance
Start instance
Reboot instance
Hibernate instance
Terminate (delete) instance

cleanup_terminate_instance.png

Snapshots (1/1) Info

Owned by me

Search

Name	Snapshot ID	Full snapshot size	Volume size	Description
snap-04c35d3bceab351e2		1.62 GiB	8 GiB	Created by Crea

Create volume from snapshot
Create image from snapshot
Copy snapshot
Launch copy duration calculator
Delete snapshot

cleanup_delete_volumes_snapshots.png

Security Groups (1) Info

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description
-	sg-0567ae2c936d1bb1c	default	vpc-05883252ecb09ac3e	default VPC secur

Successfully deleted 1 key pair

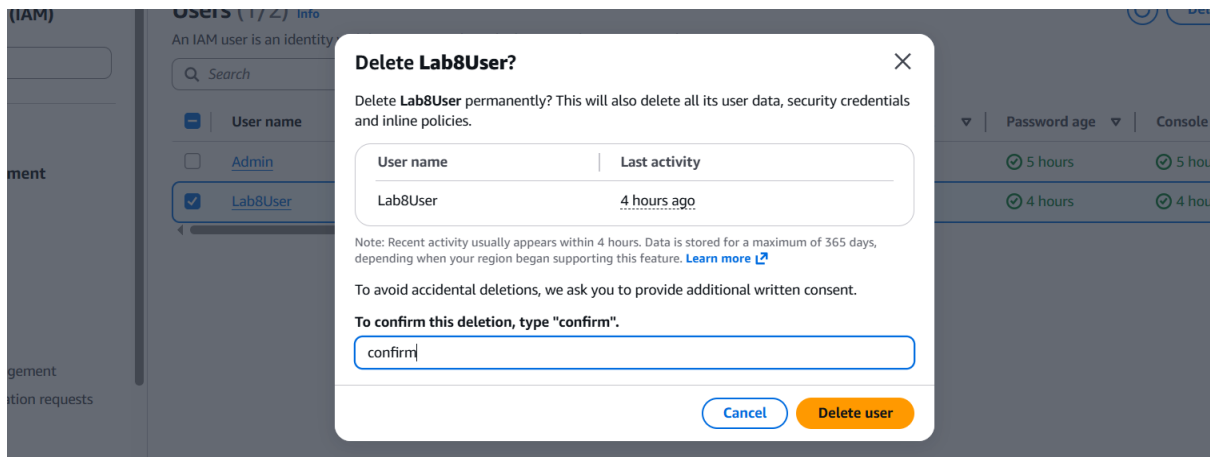
Key pairs (1) Info

Find Key Pair by attribute or tag

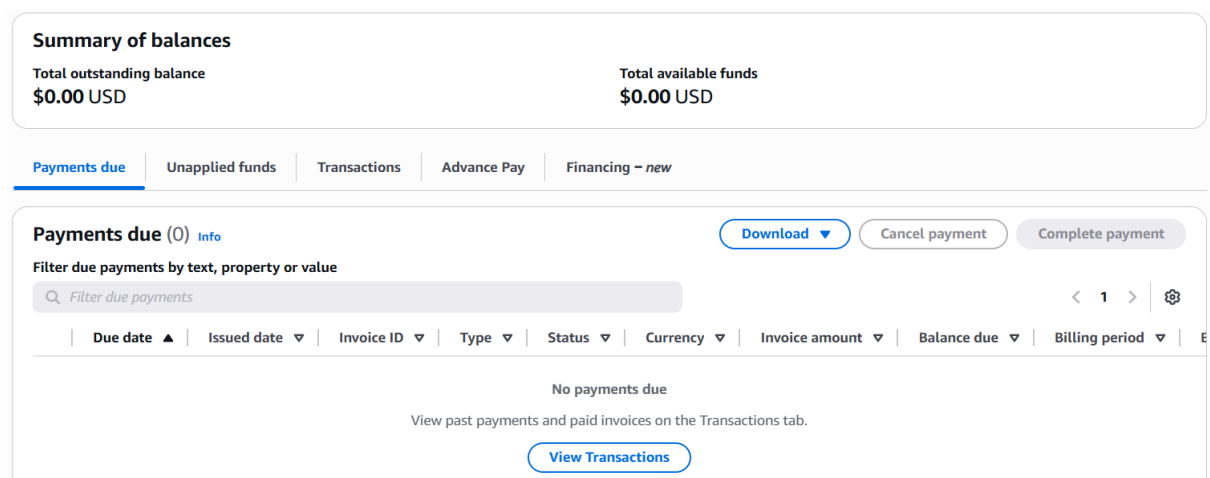
Name: Lab8Key

Name	Type	Created	Fingerprint	ID
No key pairs to display				

cleanup_delete_security_group_and_keypair.png



cleanup_iam_users_deleted.png



cleanup_summary.png